

1. Mean, Median, and Mode

Mean

The **Mean** is one of the most commonly used statistical measures to find the average value of a dataset. It is calculated by adding all the values together and dividing the result by the total number of observations. Mean helps researchers and analysts understand the central or typical value of the data. In finance and banking, mean is very useful because it gives an overall picture of customer income, loan amount, or credit behaviour. However, the mean can be affected by extremely high or extremely low values called outliers. For example, if one customer has an exceptionally large income, the mean income may become higher than the actual typical income of most customers.

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Where:

- $\sum x$ = sum of all values
- n = total number of observations

Real-World Example

Suppose five customers applying for loans have incomes of ₹40,000, ₹50,000, ₹60,000, ₹70,000, and ₹80,000. The mean income is calculated by adding all the incomes and dividing by 5. The result is ₹60,000, which represents the average earning level of customers.

Use in This Project

In this project, mean is used to calculate:

- Average customer income
- Average loan amount
- Average credit score

This helps banks understand the general financial condition of customers and evaluate loan repayment capacity.

Median

The **Median** is the middle value of a dataset when all values are arranged in ascending or descending order. If the number of observations is odd, the middle value is the median. If the number of observations is even, the median is calculated by taking the average of the two middle values. Median is an important statistical measure because it is less affected by outliers compared to the mean. Therefore, it gives a more accurate representation of the central value when the dataset contains extreme values.

Formula

For odd number of observations:

$$\text{Median Position} = \left(\frac{n + 1}{2} \right)$$

Real-World Example

Suppose customer incomes are ₹30,000, ₹40,000, ₹50,000, ₹60,000, and ₹10,00,000. Although one customer has an extremely high income, the median remains ₹50,000 because it is the middle value. This shows that median represents the typical income more effectively than the mean when outliers exist.

Use in This Project

Median is used in this project to:

- Identify the typical customer income
- Reduce the effect of extreme income values
- Compare with mean to analyse skewness in data

Banks use median income to better understand the normal earning level of customers.

Mode

The **Mode** is the value that appears most frequently in a dataset. Unlike mean and median, mode can be used for both numerical and categorical data. A dataset may have one mode, more than one mode, or no mode at all. Mode is particularly useful when identifying the most common behavior or preference among customers.

Formula

There is no mathematical formula for mode because it is simply the most repeated value in the dataset.

Real-World Example

Suppose loan terms selected by customers are:

12 months, 24 months, 24 months, 36 months, 48 months, and 24 months.

The mode is 24 months because it appears most frequently. This means most customers prefer loans with a 24-month repayment period.

Use in This Project

Mode helps identify:

- Most common loan term
- Most common credit score range
- Most frequently occurring customer behavior

This helps financial institutions understand customer preferences.

2. Variance and Standard Deviation

Variance

Variance is a statistical measure that shows how far data values are spread from the mean. It helps determine whether the data points are closely clustered around the average or widely scattered. A low variance means the values are close to the mean, while a high variance means there is large variability in the dataset.

Formula

$$\sigma^2 = \frac{\sum(x - \mu)^2}{N}$$

Where:

- x = each observation
- μ = mean
- N = total observations

Real-World Example

If all customers borrow similar loan amounts, variance will be low. However, if some customers borrow ₹50,000 while others borrow ₹20,00,000, the variance becomes very high because the loan amounts are widely spread.

Use in This Project

Variance is used to:

- Measure variability in loan amounts
- Understand financial spread among customers
- Analyse consistency in borrowing behaviour

A high variance may indicate greater financial risk.

Standard Deviation

Standard deviation is the square root of variance and measures the average distance of data values from the mean. It is easier to interpret because it uses the same units as the original data. Standard deviation is widely used in finance, economics, and machine learning to measure risk and uncertainty.

Formula

$$\sigma = \sqrt{\frac{\sum(x - \mu)^2}{N}}$$

Real-World Example

Suppose the average loan amount is ₹5,00,000. If the standard deviation is very high, it means customer loan amounts differ significantly from the average. Some customers may borrow much less, while others may borrow much more.

Use in This Project

Standard deviation helps:

- Analyse loan risk
- Measure income variability
- Detect unusual customer behavior
- Understand financial stability of applicants

Banks use standard deviation to assess financial uncertainty.

3. Random Variable

A **Random Variable** is a variable whose value depends on the outcome of a random process or uncertain event. It converts random outcomes into numerical form, making probability calculations easier. Random variables are widely used in statistics, machine learning, finance, and risk analysis.

There are two main types:

1. Discrete Random Variable
2. Continuous Random Variable

Discrete variables take specific countable values, while continuous variables can take any value within a range.

Formula Representation

$$X = \{x_1, x_2, x_3, \dots\}$$

Real-World Example

In this project, Default status is a random variable because a customer may either:

- Default on the loan
- Repay successfully

Similarly:

- Income
- Loan amount
- Credit score

are also random variables because they vary among customers.

Use in This Project

Random variables are used to:

- Build probability models
- Predict customer default

- Analyse uncertain financial outcomes
- Support decision-making in loan approval systems

4. Conditional Probability

Conditional probability measures the probability of an event occurring given that another event has already occurred. It helps analyse relationships between two events and is very important in predictive analytics and risk management.

Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A | B) \cdot P(B)$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \approx 0.46$$

$P(B)=0.65$
 $P(A \cap B)=0.30$
 $P(A | B) \approx 0.46$
 $A \cap B$ is the part of B where A also happens

Real-World Example

Suppose a bank wants to calculate:

Probability that a customer defaults given that their credit score is below 600.

If:

- 200 customers have credit score below 600
- 130 among them defaulted

Then conditional probability becomes:

$$130/200 = 0.65$$

So, there is a 65% probability of default among low-credit-score customers.

Use in This Project

Conditional probability helps:

- Identify high-risk customers
- Predict default likelihood
- Improve loan approval decisions
- Reduce financial losses for banks

5. Bayes Theorem

Bayes Theorem is a mathematical method used to update probabilities when new evidence or information becomes available. It combines prior knowledge with observed data to produce more accurate probability predictions. Bayes Theorem is very important in machine learning, artificial intelligence, banking, fraud detection, and medical diagnosis.

Formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \approx 0.68, P(B) \approx 0.25$$

$P(B)=0.25$ $P(B|A)P(A)=0.17$ $P(A|B)\sim 0.68$ Posterior = useful evidence / total evidence

Real-World Example

A bank may already know that customers with low credit scores have a higher probability of default. Using Bayes Theorem, the bank can update the probability of default after observing a customer's credit score and repayment history.

Use in This Project

Bayes Theorem is used to:

- Predict loan default probability
- Build intelligent loan risk systems
- Improve decision-making using prior customer data
- Support probabilistic financial analysis

6. Empirical Probability and Theoretical Probability

Empirical Probability

Empirical probability is based on actual observations and experimental data. It is calculated using real-world outcomes rather than assumptions.

Formula

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

Real-World Example

If:

- 1500 customers defaulted

- Total customers = 5000

Then the probability of default is:

$$1500/5000 = 0.30$$

This means the empirical probability of default is 30%.

Theoretical Probability

Theoretical probability is based on mathematical assumptions and expected outcomes rather than actual observations.

Real-World Example

The probability of getting heads when tossing a fair coin is:

$$1/2$$

This value is based on theory, not actual experiment results.

Use in This Project

Empirical probability is mainly used in this project because the analysis is based on real customer data. It helps calculate actual loan default rates and financial risk levels.

7. Poisson Distribution

Poisson Distribution is a probability distribution used to measure how many times an event occurs within a fixed interval of time, area, or space. It is mainly used for rare and independent events.

Formula

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where:

- λ = average rate of occurrence
- x = number of occurrences

Real-World Example

Suppose a bank observes an average of 5 loan defaults per day. Poisson distribution can help predict the probability of exactly 7 defaults occurring tomorrow.

Use in This Project

Poisson distribution helps:

- Estimate frequency of loan defaults
- Analyse rare financial events

- Predict operational banking risks
 - Support financial forecasting
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8. Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors are advanced concepts in linear algebra used in machine learning, artificial intelligence, and data analysis. When a matrix transforms a vector, the eigenvector remains in the same direction, while the eigenvalue tells how much the vector is stretched or compressed.

These concepts are important for understanding multidimensional data and identifying hidden patterns.

Formula

$$Av = \lambda v$$

Where:

- A = matrix
- v = eigenvector
- λ = eigenvalue

Real-World Example

In banking systems, customer data contains many variables such as:

- income
- loan amount
- credit score
- age

Eigenvectors help identify the most important directions or patterns in this data, while eigenvalues measure the importance of those patterns.

Use in This Project

Eigenvalues and eigenvectors are used in:

- Principal Component Analysis (PCA)
- Dimensionality reduction
- Feature extraction
- Identifying important financial patterns
- Simplifying large datasets while preserving important information .